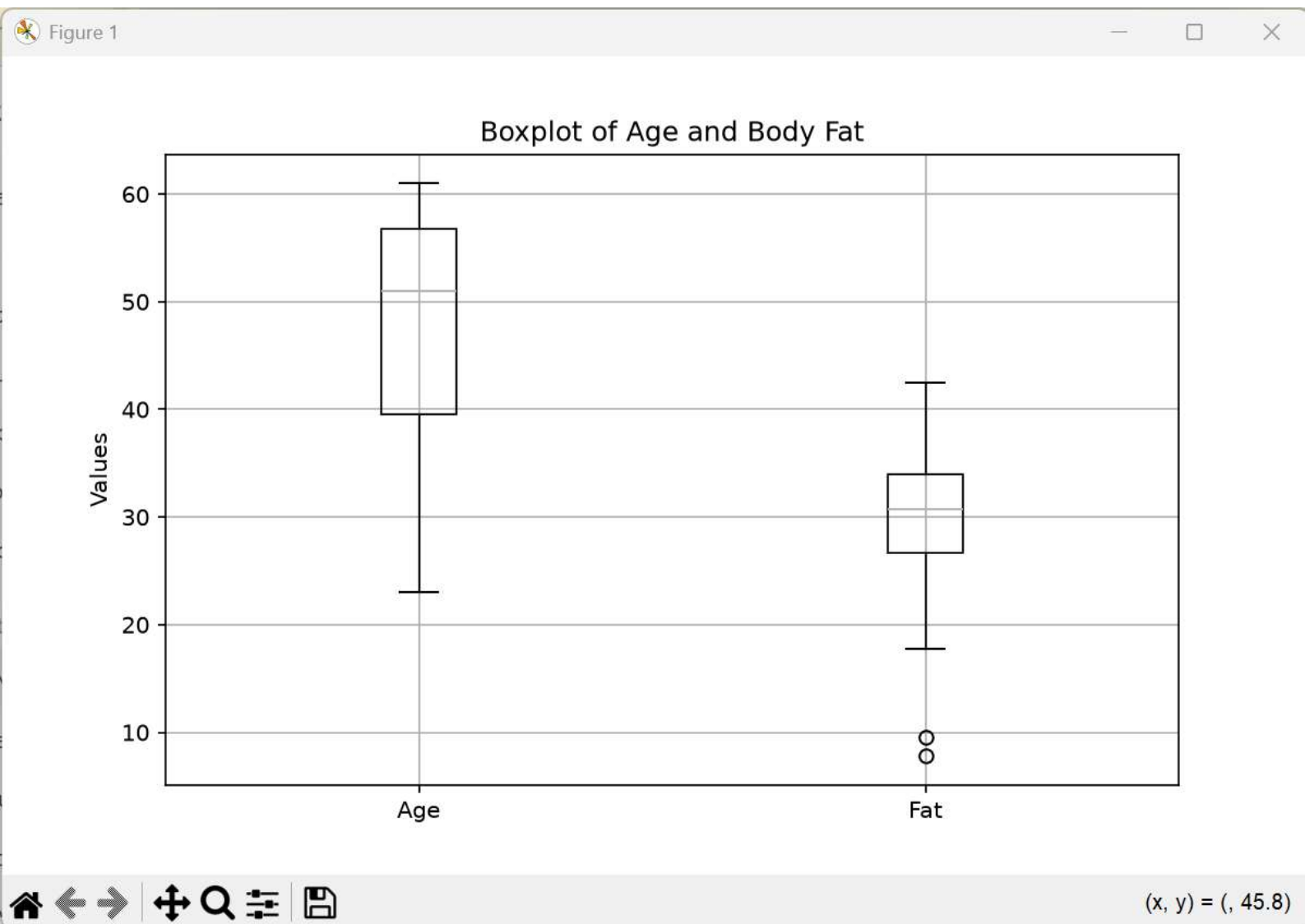




```
*IDLE Shell 3.13.13*

File Edit Shell Debug Options Window Help

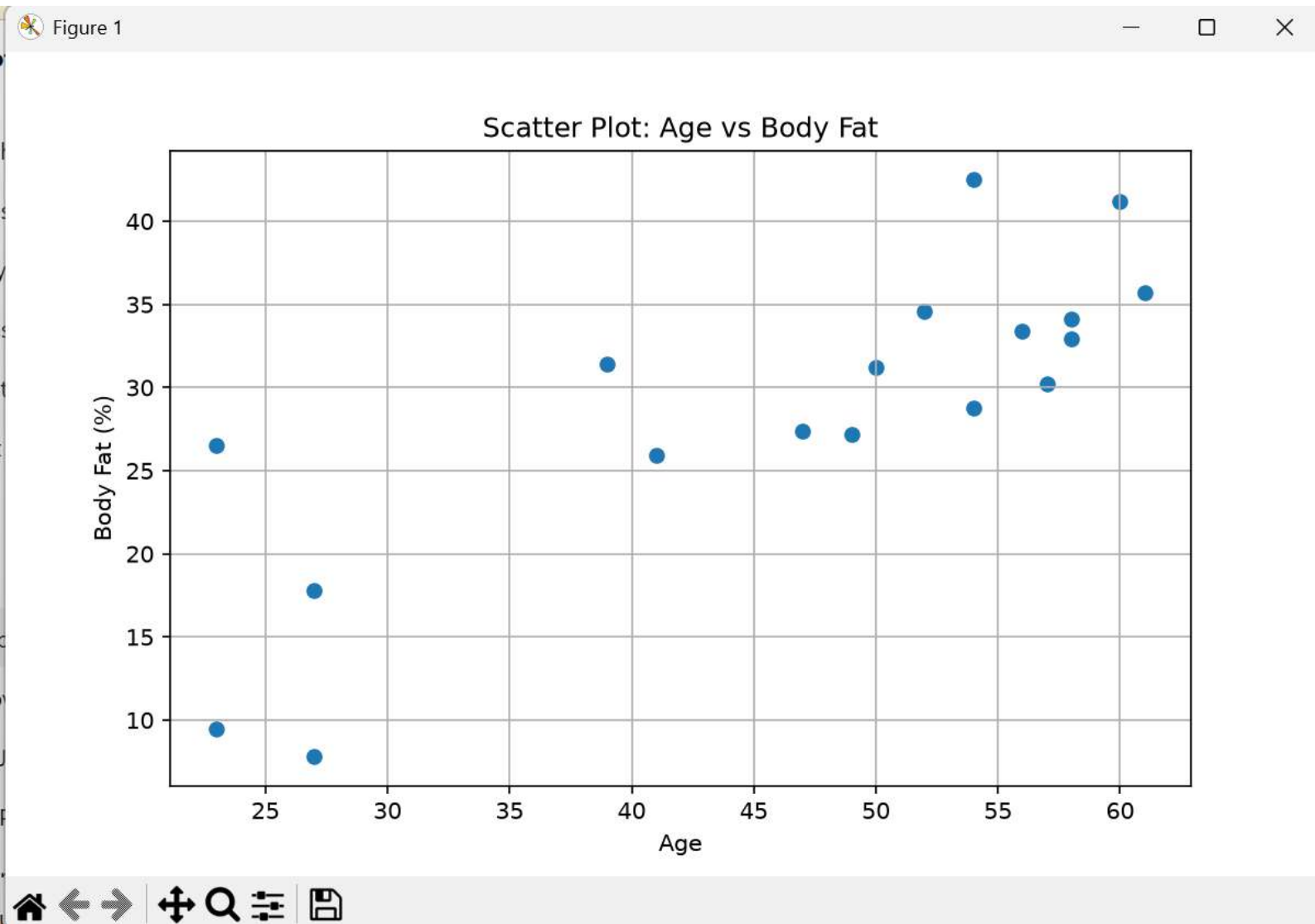
Python 3.13.13 (tags/v3.13.13:01104ce, Apr 7 2026, 19:25:48) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/Admin/Downloads/21..py =====
Statistical Analysis
-----

Mean:
Age      46.444444
Fat      28.783333
dtype: float64

Median:
Age      51.0
Fat      30.7
dtype: float64

Standard Deviation:
Age      13.218624
Fat      9.254395
dtype: float64
```



Q-Q Plot of Age

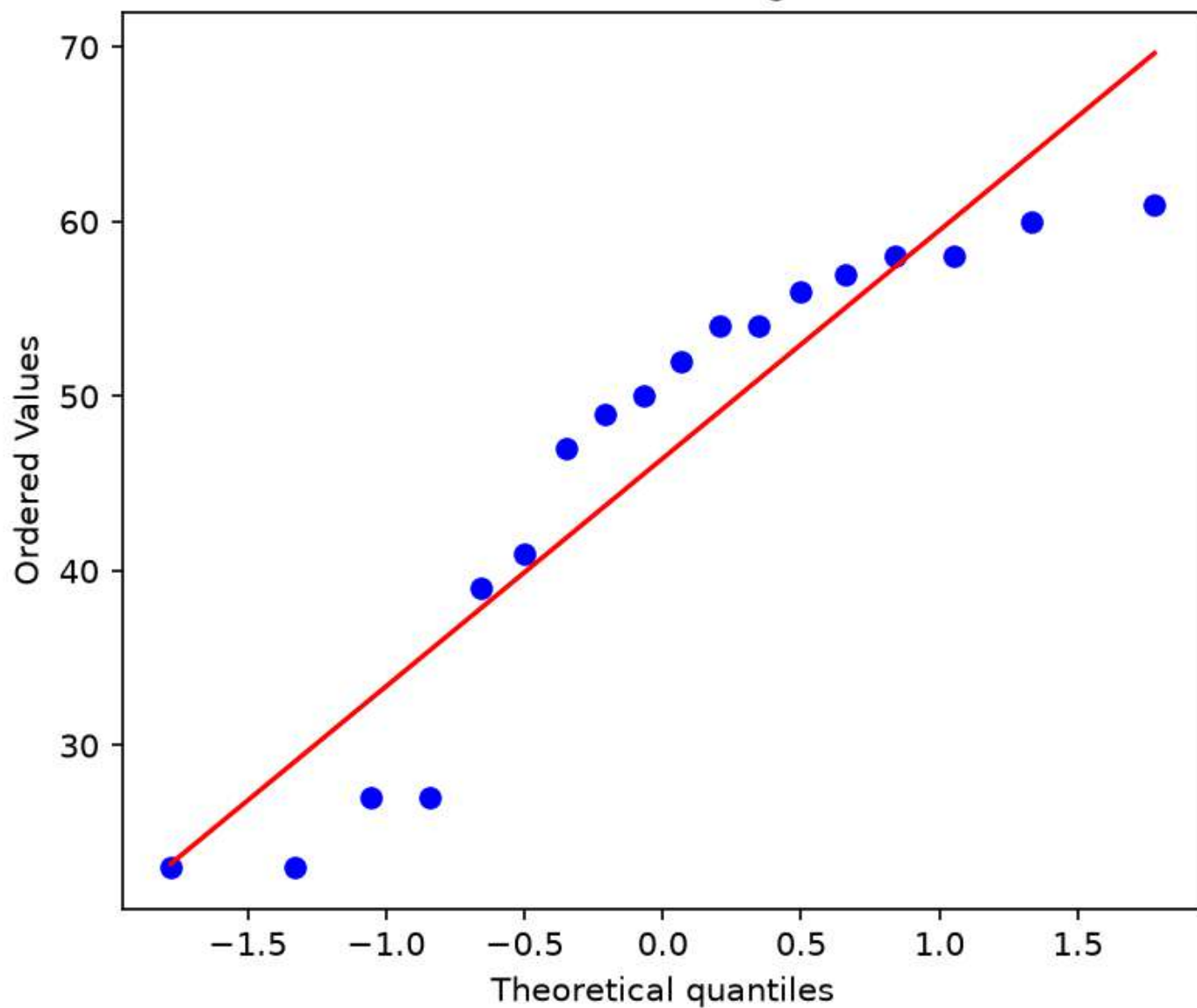
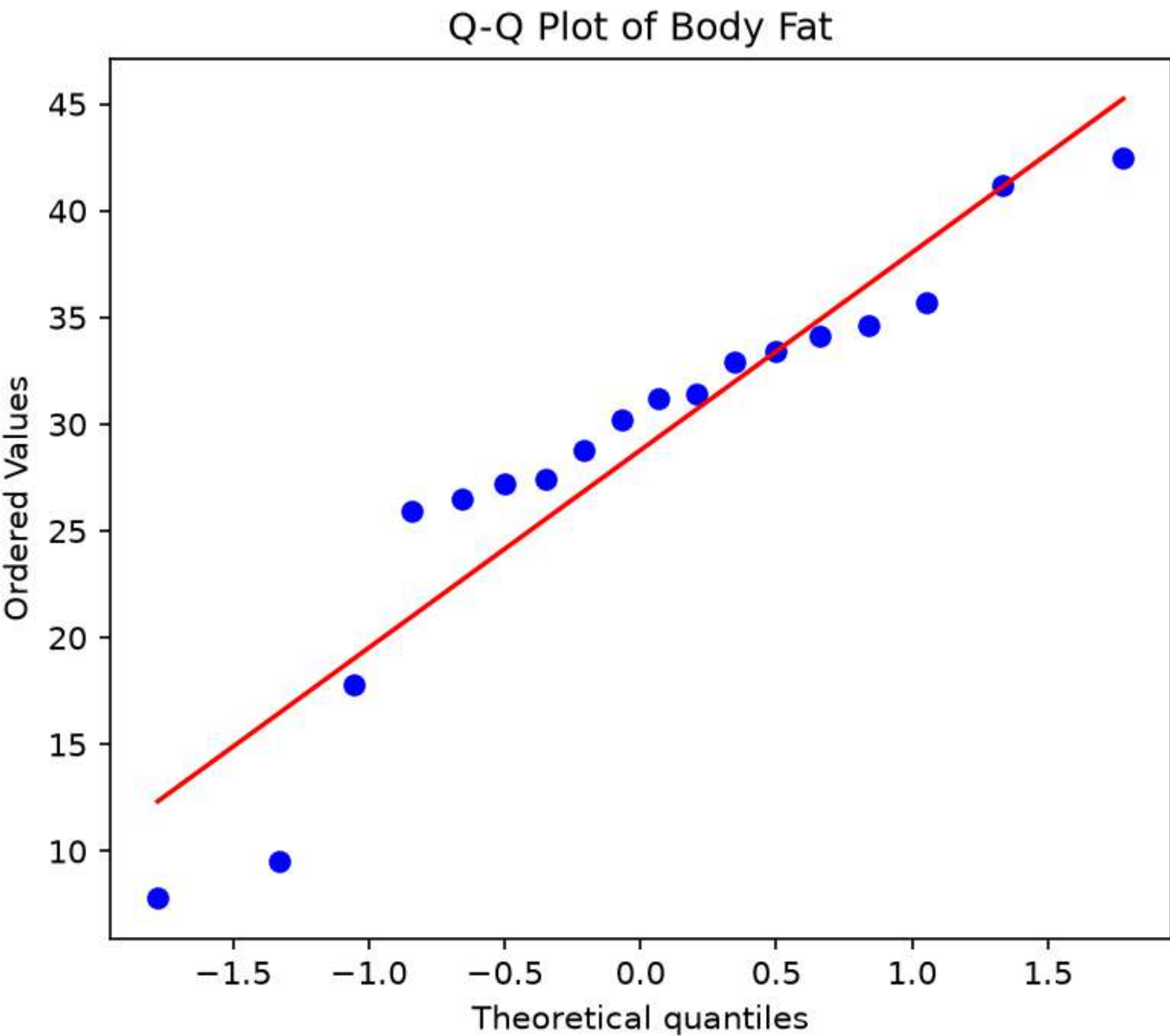


Figure 1



```
21..py - C:/Users/Admin/Downloads/21..py (3.13.13)
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt
import scipy.stats as stats

# Data from the question
age = [23, 23, 27, 27, 39, 41, 47, 49, 50,
       52, 54, 54, 56, 57, 58, 58, 60, 61]

fat = [9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,
       34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7]

# Create DataFrame
df = pd.DataFrame({
    "Age": age,
    "Fat": fat
})

# Calculate mean, median and standard deviation
print("Statistical Analysis")
print("-----")

print("\nMean:")
print(df.mean())

print("\nMedian:")
print(df.median())

print("\nStandard Deviation:")
print(df.std())

# -----
# Boxplots
# -----
plt.figure(figsize=(8, 5))
df.boxplot(column=["Age", "Fat"])
plt.title("Boxplot of Age and Body Fat")
plt.ylabel("Values")
plt.show()

# -----
```

```
21..py - C:/Users/Admin/Downloads/21..py (3.13.13)
File Edit Format Run Options Window Help

print("\nStandard Deviation:")
print(df.std())

# -----
# Boxplots
# -----
plt.figure(figsize=(8, 5))
df.boxplot(column=["Age", "Fat"])
plt.title("Boxplot of Age and Body Fat")
plt.ylabel("Values")
plt.show()

# -----
# Scatter Plot
# -----
plt.figure(figsize=(8, 5))
plt.scatter(df["Age"], df["Fat"])
plt.title("Scatter Plot: Age vs Body Fat")
plt.xlabel("Age")
plt.ylabel("Body Fat (%)")
plt.grid(True)
plt.show()

# -----
# Q-Q Plot for Age
# -----
plt.figure(figsize=(6, 5))
stats.probplot(df["Age"], dist="norm", plot=plt)
plt.title("Q-Q Plot of Age")
plt.show()

# -----
# Q-Q Plot for Body Fat
# -----
plt.figure(figsize=(6, 5))
stats.probplot(df["Fat"], dist="norm", plot=plt)
plt.title("Q-Q Plot of Body Fat")
plt.show()
```